

popquiz conservation energy

 This is a preview of the published version of the quiz

Started: Apr 29 at 8:04pm

Quiz Instructions

only one attempt



Question 1 17 pts

A large household air conditioner may consume 15.0kW of power. What is the cost in dollars of operating this air conditioner 3.00h per day for 30.0d if the cost of electricity is \$0.20 per kWh?

KWh is a unit for energy.

energy(KWh)= power(KW) x time (hours) and 1KW = 1000 W

note: if energy is in joules. power is in W and time in seconds.



\$25



about \$50



about \$30



about \$270



Question 2 17 pts

The empire state building is about 400m. An elephant takes a wrong step and falls. Before it hits the ground, its speed is ?

(use the conservation of energy)

Give the answer is mph . 1m/s is 2.2mph (round).



about 90 mph



about 100mph



about 20mph



about 200 mph



Question 3 17 pts

Building on the previous question. The elephant has a mass of 5000 kg. Use your previous answer in m/s (not mph). The elephant hits the floor and stops (obviously). What is its change in momentum (unit is kg m/s)



5000 kg m/s



445,000 kg m/s



220,000 kg m/s



150,000 kg m/s



Question 4 17 pts

Building on the previous question. The change in momentum is the impulse $F \times \text{time}$. If the time of impact is 1 ms (0.001s) , what is the force acting on the elephant?

(remember: impulse = change in momentum . impulse = $F \times t$)



445,000,000 N (89,000,000 lbs)



150,000,000 (30,000,000 lbs)



220,000,000N (44,000,000 lbs)



5,000,000N (1,000,000 lbs)



Question 5 17 pts

We produce (on average) the same amount of energy per second than a 100 W light bulb (incandescent). So our power is about 100W .

Find the energy produced by a person in one day in Wh (multiply watts by number of hours). Then convert to food calories (Cal). 1 Wh is 1 Cal

This is the amount of Cal (on average) we lose per day.

☐

24 C

☐

140 Cal

☐

2400 Cal

☐

1000 Cal



Question 6 17 pts

Your mass is 80kg and you climb the stairs of the empire state building (400m about).

Find the amount of work you have to do again gravity (which is the PEG). Your answer is in joules.

1 Cal = 4000 J. How much Cal for this effort?

Isn't it demoralizing?

☐

50 Cal

☐

400 Cal

☐

80 Cal

☐

320,000 Cal

Not saved

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